

# Bioinformatics session

4th annual workshop on  
bioinformatics and variant  
interpretation in InPreD

[https://inpred.github.io/26-09\\_bioinfo\\_ws/](https://inpred.github.io/26-09_bioinfo_ws/)



## 2. Containerization



## A brief history

|             |                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                                                                                                                                                                                                               |
| <b>1979</b> | <code>chroot</code> system call (changing the root directory of a process and its children to a new location in the filesystem) in Uni V7 considered as beginning of process isolation                        |
| <b>2000</b> | FreeBSD Jails allows administrators to partition FreeBSD computer system into several independent, smaller systems ( <i>jails</i> ) – with the ability to assign IP address for each system and configuration |
| <b>2001</b> | Linux VServer is jail mechanism that can partition resources, similar to FreeBSD Jails                                                                                                                        |

|             |                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                                                                                                                                                                                                                 |
| <b>2004</b> | First public beta of Solaris Containers released - combines system resource controls and boundary separation provided by zones                                                                                  |
| <b>2005</b> | Open Virtuozzo is operating system-level virtualization technology for Linux which uses patched Linux kernel for virtualization, isolation, resource management and checkpointing                               |
| <b>2006</b> | Process Containers, launched by Google, was designed for limiting, accounting and isolating resource usage of collection of processes - renamed to <i>Control Groups (cgroups)</i> and merged into Linux kernel |
| <b>2008</b> | LinuX Containers (LXC) was first, most complete implementation of Linux container manager - implemented using cgroups and Linux namespaces                                                                      |



|             |                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|             |                                                                                                                                             |
| <b>2011</b> | CloudFoundry started Warden can isolate environments on any operating system running as daemon and providing API for container management   |
| <b>2013</b> | Let Me Contain That For You (LMCTFY) kicked off as open-source version of Google's container stack - providing Linux application containers |
| <b>2013</b> | Docker emerged                                                                                                                              |
| <b>2016</b> | Singularity (later Apptainer) was released                                                                                                  |

# Docker

## What is it? 🔍

- containerization technology to package application and dependencies into a container
- the container image can be shipped and run consistently across different computing environments

Which benefits does it provide?



1. Consistency
2. Isolated environments
3. Portability
4. Efficiency



## How is it different from virtual machines?

- containers use and share host OS's kernel making them more lightweight and efficient
- virtual machines emulate entire physical machine, including OS making it possible to run multiple OS instances on single physical machine



Application A

Application A

Operating System A

Docker

Virtual Machine Hypervisor

Operating System B

Infrastructure

## Key Concepts 🔑

1. **Image:** standardized package that includes all files, binaries, libraries, and configurations to run container
2. **Container:** running instance of image providing isolated runtime environment
3. **Dockerfile:** instructions on how to build image
4. **Docker Hub:** public registry where developers can share and access pre-built images

## Key Components 🔑

1. **Docker Engine:** core, open-source technology that builds and runs containers
2. **Docker Client:** command-line tool that sends instructions to the Docker Daemon using REST APIs
3. **Docker Daemon ( `dockerd` ):** receives and processes API requests and calls container runtime
4. **Container Runtime ( `containerd` ):** turns static container image into running, isolated application on host OS; industry standard

# Let's explore! 🌐

Start by going to [https://github.com/InPreD/26-09\\_bioinfo\\_ws\\_docker\\_and\\_ci](https://github.com/InPreD/26-09_bioinfo_ws_docker_and_ci) and create a fork

The screenshot shows the GitHub interface for the repository '26-09\_bioinfo\_ws\_docker\_and\_ci'. At the top, there are buttons for 'Edit Pins', 'Watch' (0), 'Fork' (0), and 'Star' (0). Below these, a dropdown menu is open, showing the option '+ Create a new fork' highlighted with a red rectangle. The repository is owned by 'marrip' and has 1 branch and 0 tags. The main content area shows a list of files and their commit history:

| File           | Commit Message                  | Time Ago      |
|----------------|---------------------------------|---------------|
| .devcontainer  | chore: add devcontainer setup   | 9 minutes ago |
| src/greeter    | feat: add simple python project | 8 minutes ago |
| .gitignore     | chore: add devcontainer setup   | 9 minutes ago |
| LICENSE        | Initial commit                  | 1 hour ago    |
| README.md      | Initial commit                  | 1 hour ago    |
| pyproject.toml | feat: add simple python project | 8 minutes ago |

Below the file list, there is a section for the README, which includes the title '26-09\_bioinfo\_ws\_docker\_and\_ci' and the description 'codespace template for docker and ci workshop'. On the right side, there are links to 'workshop', 'Readme', 'AGPL-3.0 license', 'Activity', 'Custom properties', '0 stars', '0 watching', '0 forks', 'Audit log', 'Report repository', 'Releases', and 'Packages'.

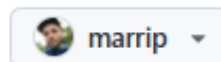
# Create your own fork by clicking on **Create fork**

## Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

*Required fields are marked with an asterisk (\*).*

Owner \*



Repository name \*

/ 26-09\_bioinfo\_ws\_docker\_

✓ 26-09\_bioinfo\_ws\_docker\_and\_ci is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

codespace template for docker and ci workshop

45 / 350 characters

☒ Copy the `main` branch only

Contribute back to InPreD/26-09\_bioinfo\_ws\_docker\_and\_ci by adding your own branch. [Learn more.](#)

You are creating a fork in your personal account.

**Create fork**



In the forked repository, navigate to **Code** > **Codespaces** > **Create codespace on main**

The screenshot shows the GitHub interface for a repository named '26-09\_bioinfo\_ws\_docker\_and\_ci'. The repository is public and has 1 branch and 0 tags. The 'Code' button is highlighted in green. A dropdown menu is open, showing 'Local' and 'Codespaces' options. The 'Codespaces' option is selected, and a sub-menu is displayed with the text 'No codespaces' and 'You don't have any codespaces with this repository checked out'. A green button labeled 'Create codespace on main' is highlighted with a red box. Below this button is a link 'Learn more about codespaces...'. The repository description is 'codespace template for docker and ci workshop'. The 'About' section on the right lists the license as AGPL-3.0 and shows 0 stars, 0 watching, and 0 forks. The 'Releases' and 'Packages' sections also show no published items.

26-09\_bioinfo\_ws\_docker\_and\_ci Public

Edit Pins Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file

Add file

Code

Local Codespaces

Codespaces  
Your workspaces in the cloud

No codespaces  
You don't have any codespaces with this repository checked out

Create codespace on main

Learn more about codespaces...

Codespace usage for this repository is paid for by marrip.

26-09\_bioinfo\_ws\_docker\_and\_ci

codespace template for docker and ci workshop

About  
codespace template for docker and ci workshop

Readme  
AGPL-3.0 license  
Activity  
Custom properties  
0 stars  
0 watching  
0 forks  
Audit log  
Report repository

Releases  
No releases published  
Create a new release

Packages  
No packages published  
Publish your first package

Run the following commands:

```
# check for running docker containers
$ docker ps
# check for existing images
$ docker images
# pull image
$ docker pull ubuntu:26.04
# run image
$ docker run ubuntu:26.04
```

Run an interactive container:

```
# start interactive container
$ docker run -it ubuntu:26.04
# print container os version
@ cat /etc/lsb-release
# exit container
@ exit
# print os version
$ lsb_release -a
```

Instruct docker to remove containers after use:

```
# check for all docker containers
$ docker ps -a
# remove stopped container
$ docker rm <container hash>
# run docker with --rm flag
$ docker run -it --rm ubuntu:26.04
# exit container
@ exit
# check for all docker containers
$ docker ps -a
```

## Remove the docker image

```
# remove docker image  
$ docker rmi ubuntu:26.04  
# alternatively  
$ docker rmi <container image hash>
```

## Building an image

We start off by creating a `Dockerfile` in the root directory of our repository. We add the following to our file:

```
FROM python:3.14-slim-trixie
RUN echo 'Hello world' > greetings.txt
```

And then we build and run our docker image

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker images
$ docker run --rm greeter:test cat greetings.txt
```

### **3. Continuous Integration (CI)**





